

## Solving Absolute Value Linear Inequalities

Algebra II | Unit 2 | Day 7 | TEK A2.6.F

**Objective:** I can solve absolute value linear inequalities using the graphing method and the algebraic (compound inequality) method.

### Vocabulary

**Compound inequality** — two inequalities joined by \_\_\_\_\_ or \_\_\_\_\_.

**"Or" statement** — true if EITHER part is true - used for \_\_\_\_\_ or \_\_\_\_\_.

**"And" statement** — true only if BOTH parts are true - used for \_\_\_\_\_ or \_\_\_\_\_.

### Part 1 — The Graphing Method

Compare  $y1 = |expression|$  to  $y2 = constant$ . Where  $y1$  is BELOW  $y2$ , that is the "less than" solution; where  $y1$  is ABOVE  $y2$ , that is the "greater than" solution.

### Part 2 — The Algebraic Method

**Example 1 ( $<$ , uses AND).**  $|2x + 3| < 5$

$$-5 < 2x + 3 < \underline{\hspace{2cm}} \rightarrow -8 < 2x < 2 \rightarrow \underline{\hspace{2cm}}$$

**Example 2 ( $\geq$ , uses OR).**  $|2x - 3| + 4 \geq 9$

Isolate:  $|2x - 3| \geq \underline{\hspace{2cm}}$

$$2x - 3 \leq -5 \text{ or } 2x - 3 \geq \underline{\hspace{2cm}} \rightarrow \underline{\hspace{2cm}}$$

### Part 3 — You Try

1.  $|x + 4| \leq 6$  \_\_\_\_\_

2.  $|3x - 1| > 8$  \_\_\_\_\_

3.  $|x - 5| + 2 \geq 9$  \_\_\_\_\_

### Practice

1.  $|2x| < 10$  \_\_\_\_\_

2.  $|x + 1| \geq 3$  \_\_\_\_\_

3.  $3|x - 2| < 12$  \_\_\_\_\_

4. Is  $x = 0$  a solution to  $|x - 3| > 5$ ? \_\_\_\_\_

### Exit Ticket

Solve  $|3x + 2| \geq 7$ , and state whether it is an "and" or "or" compound inequality.

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